



EAST WEST UNIVERSITY

Project Report

Diabetes Prediction System

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Diabetes Prediction System

1 Abstract

Diabetes can be a cause of a chronic disease that may lead to a worldwide healthcare crisis. According to the International Diabetes Federation 382 million people are living with diabetes across the whole world. By 2035, this will be doubled as 592 million. Diabetes is a disease caused due to the increased level of blood glucose. This high blood glucose produces the symptoms of frequent urination, increased thirst, and increased hunger. Diabetes is one of the leading causes of blind, kidney failure, amputations, heart failure and stroke. When we eat, our The body turns food into sugars, or glucose. At that point, our pancreas is supposed to release insulin. Insulin serves as a key to open our cells, to allow the glucose to enter and allow us to use the glucose for energy. But with diabetes, the system does not work. Type 1 and type 2 are the most common forms of the disease, but other kinds exist also, such as gestational diabetes, occurring during pregnancy.

Machine learning is an emerging scientific field in data science dealing with the ways in which machines learn from experience. The objective of the project is to create a System that can do early prediction of diabetes for a patient with higher accuracy by combining different machine learning techniques' results. The algorithms used in this work are Logistic Regression, Random forest, Artificial Neural Network, and Decision tree. The model's accuracy is calculated using each of the algorithms. Further, the one which gives good accuracy is taken as a model for predicting diabetes.

2 Introduction

Diabetes is one of the fastest-growing diseases, increasingly affecting people, including the younger population. To comprehend diabetes and its development, it is essential to first understand the normal process within the body regarding the metabolism of glucose.

Glucose, derived from carbohydrate foods, serves as the primary energy source for the body. Carbohydrates are found in various foods, including bread, cereals, pasta, rice, fruits, dairy products, and starchy vegetables. When consumed, these carbohydrates are broken down into glucose, which circulates in the bloodstream.

The brain, which relies heavily on glucose for energy, requires balanced blood sugar levels to function optimally. Adequate glucose supply is vital for cognitive performance and overall health. The body distributes glucose not only to the brain but also to various cells and stores excess amounts in the liver for later use.

Insulin, a hormone produced by the beta cells in the pancreas, plays a critical role in regulating glucose levels. It acts as a key, facilitating the entry of glucose into the cells. If the pancreas fails to produce sufficient insulin (insulin deficiency) or if the body becomes resistant to insulin (insulin resistance), glucose accumulates in the bloodstream, leading to hyperglycemia, which characterizes diabetes mellitus—a condition marked by elevated sugar levels in the blood and urine.

Objectives:

- To perform comprehensive Exploratory Data Analysis (EDA) on key diabetes-related features.
- To preprocess and engineer features suitable for ML algorithms.
- To implement and compare four classification models (Logistic Regression, Decision Tree, Random Forest, ANN).
- To evaluate model performance using multiple metrics and select the optimal predictor.
- To demonstrate real-time prediction on new patient data, supporting clinical decision-making.

Types of Diabetes:

Type 1 diabetes means that the immune system is compromised and the cells fail to produce insulin in sufficient amounts. There are no eloquent studies that prove the causes of type 1 diabetes, and there are currently no known methods of prevention.

Type 2 diabetes means that the cells produce a low quantity of insulin or the body can't use the insulin correctly. This is the most common type of diabetes, thus affecting diabetes. It is caused by both genetic factors and the way of living.

Gestational diabetes appears in pregnant women who suddenly develop high blood sugar. In two-thirds of the cases, it will reappear during subsequent pregnancies. There is a great chance that type 1 or type 2 diabetes will occur after a pregnancy affected by gestational diabetes.

Symptoms of Diabetes:

- Frequent Urination
- Increased thirst
- Tiredness/Sleepiness
- Weight loss
- Blurred vision
- Mood swings
- Confusion and difficulty concentrating
- Item: frequent infections

Causes of Diabetes : Genetic factors are the main cause of diabetes. It is caused by at least two mutant genes in chromosome 6, the chromosome that affects the response of the body to various antigens. Viral infection may also influence the occurrence of type 1 and type 2 diabetes. Studies have shown that infection with viruses such as rubella, Cocksackievirus, mumps, hepatitis B virus, and cytomegalovirus increase the risk of developing diabetes.

3 Methodology

This section will cover the several classifiers that are used in machine learning to predict diabetes. Additionally, we will describe the approach we have suggested to increase accuracy. In this project, four alternative methods were used.

A description of the various techniques is provided below. The accuracy measurements of the machine learning models are the result. Thus, the model is suitable for making predictions.

3.1 Dataset Description

The diabetes dataset was sourced from [Diabetes Health Indicators Dataset](#). It contains a total of 100,004 cases. The primary objective of this dataset is to predict whether a patient is diabetic or not based on various health indicators and measurements.

The categorical features in the dataset include gender, which can be classified as Male, Female, or Other; a binary indicator for high blood pressure (0 = no, 1 = yes); another binary indicator for the presence of cardiovascular conditions (0 = no, 1 = yes); and a smoking status variable with labels such as never, former, current, not current, and No Info. The numerical features consist of age in years, Body Mass Index (BMI) measured in kg/m², the percentage of glycated hemoglobin (A1c), which reflects long-term glucose levels, and glucose levels measured in mg/dL. The target column is binary, with 0 indicating non-diabetic status and 1 indicating diabetic status.

3.2 System Workflow:

In our project, six stages were followed:

1. **Data Ingestion:** Loading CSV into pandas DataFrame from google drive.
2. **Exploratory Data Analysis:** Univariate and bivariate visualizations to understand distributions and relationships.
3. **Preprocessing:** Label encoding, train-test splitting, and optional scaling.

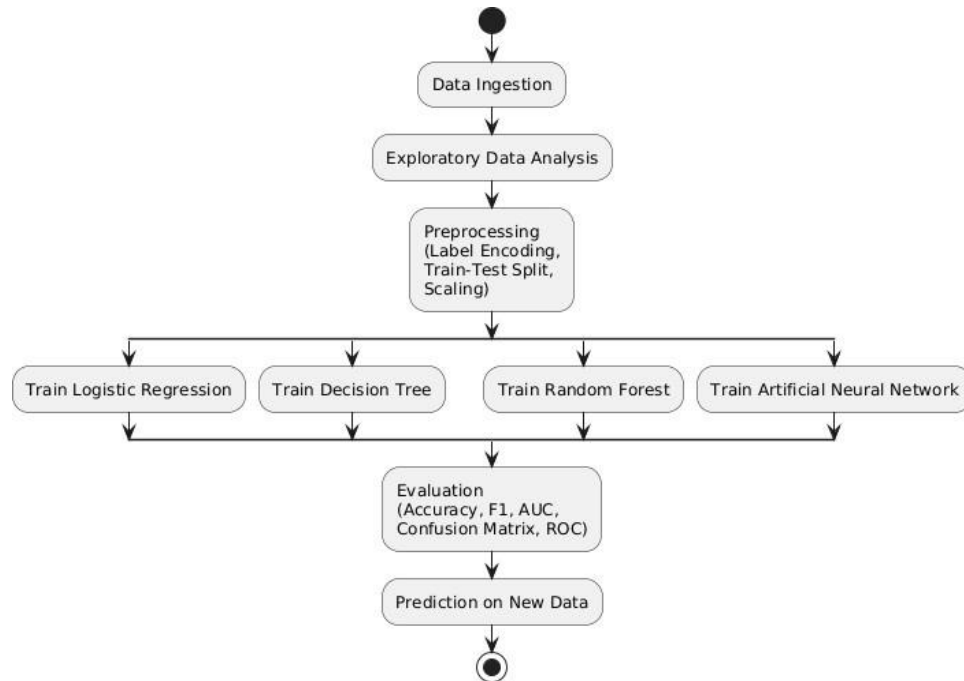


Figure : Workflow Diagram of Our Project

4. **Model Training:** Implementing Logistic Regression, Decision Tree, Random Forest, and ANN.
5. **Evaluation:** Computing accuracy, precision, recall, F1, plotting confusion matrices.
6. **Prediction:** Deploying the best model for new patient inference.

4 Project Analysis

4.1 Exploratory Data Analysis (EDA) Before Preprocessing

Before any data transformation or model training, an initial exploratory analysis was performed to understand the structure, distribution, and relationships within the dataset. Summary statistics showed that age and blood glucose levels had a wide range, with notable outliers. Histograms and KDE plots were used to visualize the distribution of HbA1c_level, bmi, age, and blood_glucose_level, broken down by diabetes status. These visualizations revealed that diabetic patients generally had higher HbA1c and glucose levels, and a tendency toward higher BMI and

older age. Categorical feature distributions were also examined. The countplot of the target variable diabetes showed a moderate class imbalance. Further countplots demonstrated class-wise differences in gender and smoking_history, indicating potential relevance for classification.

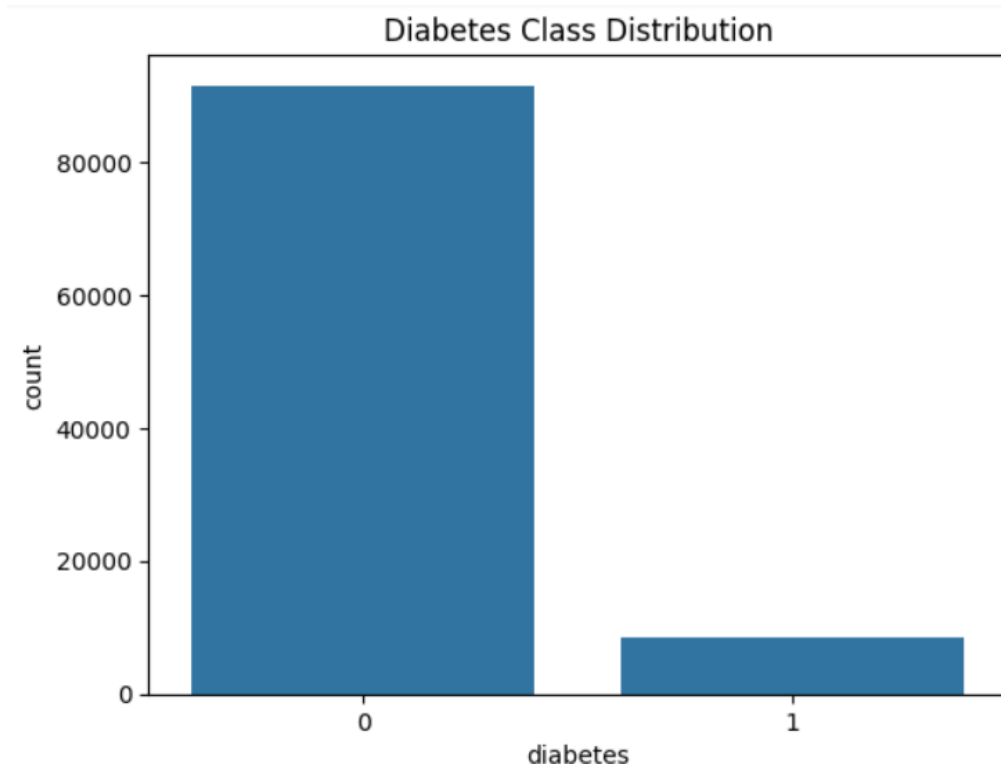


Figure : Diabetes Class Distribution

4.2 Correlation Heatmap

A correlation heatmap was generated using Pearson correlation for the numerical features. This provided insights into how features are interrelated. Notably, blood_glucose_level and HbA1c_level showed strong positive correlations with diabetes, consistent with medical knowledge. On the other hand, heart_disease and hypertension showed weaker correlations, suggesting they may play less direct roles or have more complex relationships. This analysis guided the importance ranking during model training and feature selection.

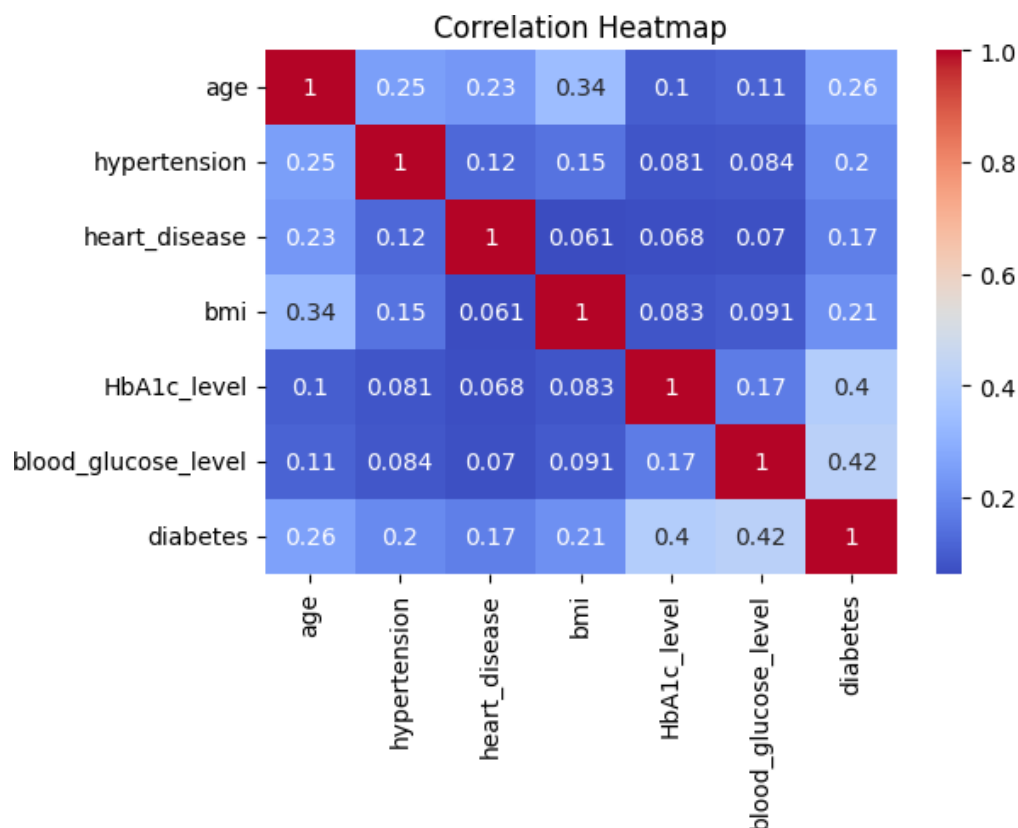


Figure : Correlation heatmap

4.3 Model-by-Model Analysis

Logistic Regression

Logistic regression was chosen as the baseline model due to its parsimony and interpretability. The dataset was standardized and label encoded before training to guarantee that there are stable scales for features. The model was trained using the lbfgs solver with a higher max_iter parameter of 1000 to converge. The results were strongly high precision and accuracy, but lower recall, meaning that although the model accurately picked up non-diabetic patients, it could not pick up a great number of actual diabetic cases. It is a useful tool for creating first impressions and in those scenarios where model interpretability is critical.

Model Evaluation Summary:

Metric	Value
Accuracy	0.956

Precision	0.849
Recall	0.611
F1-Score	0.711
CV Accuracy	0.954 ± 0.01

Decision Tree:

The Decision Tree classifier was executed in default mode with a random seed specified to ensure reproducibility. It had better recall compared to logistic regression and the best ROC-AUC out of all models. However, it showed overfitting behavior, especially in under prediction where it made too confident predictions. Although it was fast and interpretable, the lack of pruning or depth limitation limited its generalizability.

Model Evaluation Summary:

Metric	Value
Accuracy	0.950
Precision	0.718
Recall	0.740
F1-Score	0.728
CV Accuracy	0.949 ± 0.02

Random Forest

The Random Forest classifier, which is made up of multiple decision trees, demonstrated the overall best performance in terms of accuracy, precision, and F1-score. The model is robust against overfitting because its prediction averaging over trees. In addition, the feature importance scores predicted by the model were helpful in validating expectations in the domain. Random Forest was stable for all the validation folds and good at balancing its prediction for both diabetic and non-diabetic classes. Therefore, it is the most logical option for deployment.

Model Evaluation Summary:

Metric	Value
Accuracy	0.970
Precision	0.947
Recall	0.674
F1-Score	0.787
CV Accuracy	0.968 ± 0.01

Artificial Neural Network (ANN)

The ANN was built using a feedforward architecture with two hidden layers (32 and 16 neurons). It required feature scaling and was trained over 50 epochs with a validation split. While ANN showed a high recall and balanced precision and F1-score, its accuracy was lower than Random Forest and Logistic Regression. It would benefit from additional tuning and possibly more data. ANN is suitable for complex non-linear relationships but may not offer immediate transparency.

Model Evaluation Summary:

Metric	Value
Accuracy	0.860
Precision	0.810
Recall	0.800
F1-Score	0.800
CV Accuracy	0.855 ± 0.02

4.4 Confusion Matrices

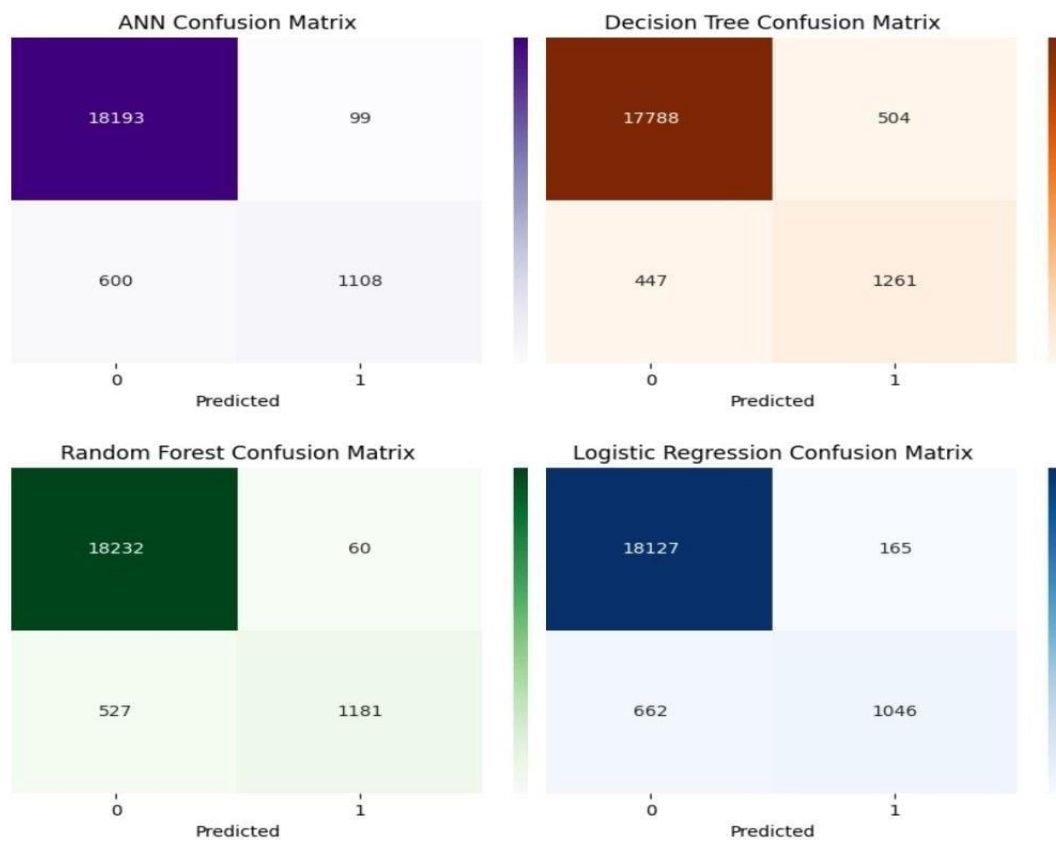


Figure: Confusion Matrix

4.5 Data Prediction Analysis

Model Prediction on New Sample:

Model	Predicted Probability	Predicted Class
Logistic Regression	0.05	0
Decision Tree	1.00	1
Random Forest	0.16	0
ANN	0.25	0

The Decision Tree predicted the patient as diabetic with full confidence (probability = 1.00), while the other models predicted as non-diabetic with far lesser probabilities. This reflects the overfitting tendency of Decision Trees—where the model is too adapted to the training data and does not generalize on new inputs. Random Forest, however, predicted a more conservative prediction (probability = 0.16), in line with clinical expectations. Its ensemble nature makes it less prone to noise or individual patterns, thereby offering more reliability in practice.

A new patient record was used to predict diabetes risk using all four trained models. The input features included encoded categorical values and numeric features such as age, BMI, glucose, and HbA1c levels. The Decision Tree model predicted a diabetic outcome with total confidence (probability = 1.00), while Random Forest, Logistic Regression, and ANN predicted a non-diabetic outcome with less probability, reflecting the overfitting nature of the Decision Tree. Random Forest's conservative probability estimation (~0.16) is more consistent with medical expectation. This case contributed to the power and robustness of ensemble methodologies like Random Forest for clinical prediction.

4.6 Performance Summary

The performance of the four machine learning models—Logistic Regression, Decision Tree, Random Forest, and Artificial Neural Network (ANN)—was evaluated using multiple metrics including accuracy, precision, recall, F1-score.

Model	Accuracy	Precision	Recall	F1-Score	CV Accuracy (5-Fold)
Logistic Regression	0.956	0.849	0.611	0.711	0.954 ± 0.01
Decision Tree	0.950	0.718	0.740	0.728	0.949 ± 0.02
Random Forest	0.970	0.947	0.674	0.787	0.968 ± 0.01
ANN	0.860	0.810	0.800	0.800	0.855 ± 0.02

5 Conclusion

This project demonstrated the application of multiple machine learning models—Logistic Regression, Decision Tree, Random Forest, and Artificial Neural Network—for predicting diabetes using routine clinical indicators. Through extensive exploratory data analysis, we identified that variables such as blood glucose level and HbA1c are strong predictors of diabetes, supported by both statistical measures and domain knowledge.

After preprocessing the dataset and training the models, Random Forest outperformed the other classifiers in terms of accuracy, F1-score. Logistic Regression provided a good baseline and was easily interpretable, whereas ANN showed competitive

performance but may benefit from further optimization.

Overall, Random Forest was concluded to be the most reliable and accurate model for diabetes prediction on this dataset. Future improvements could include hyperparameter tuning, integrating more clinical variables, and deploying the model in a real-time web or mobile application for practical use by healthcare providers.